

LIM SHAO KIAT

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SUMMARY

Data Scientist with 3 years building and shipping production ML systems at Micron Semiconductor — computer vision, predictive maintenance, and IIoT pipelines. Owned the full stack: problem framing, GCP architecture, GKE deployment, and the Angular dashboards engineers use daily. Fluent in Python, Docker, and applied GenAI, with a track record of delivering systems that manufacturing teams actually trust.

EXPERIENCE

Micron Semiconductor Asia, Data Scientist II, Singapore

Jul 2023 – Present

AI Solution Delivery & Leadership

- Led a 6-member cross-functional team to architect and deploy a production-grade computer vision platform for photolithography defect detection. Owned problem framing, system architecture, stakeholder alignment, and operational rollout; built and deployed containerised object & anomaly detection services on GKE, enabling real-time inference on a high-compliance semiconductor process currently in late-stage validation.
- Published internal technical papers presented at Data Science Technical Seminar, including:
 - *A Statistical Approach to Accurate Lifetime Prediction of Dry Etch Electrostatic Chuck*
 - *Anomaly Detection in Photo Track Coater Dispense through Video Analytics*
 - *Synthetic Data Augmentation for Robust Photo Coating Anomaly Detection*

Applied AI/ML Systems Engineering

- **User-Centric Labeling Platform:** Built a self-serve annotation tool using **Streamlit & Label Studio**, enabling engineers to label and validate defects independently; eliminated data scientist dependency and accelerated retraining cycles.
- **Synthetic Rare-Defect Generation (GenAI Augmentation):** Leveraged image-generation LLMs (**Nano Banana**) to create realistic synthetic rare-defect samples on the Photolithography process, mitigating extreme class imbalance and improving anomaly detection robustness; shortened model development cycles while increasing production readiness and generalisation performance.
- **Predictive Maintenance Platform (Time-Series ML):** Designed and implemented end-to-end time-series models from scratch (EDA, signal analysis, feature engineering, lag/rolling statistical features, degradation trend modeling), developing supervised learning pipelines to predict optimal component replacement windows. Productionised as containerised batch inference on GCP Cloud Run, currently in staged production validation with manufacturing engineers; model predicts **10%+ consumable lifetime extension**, with full rollout projected at **~USD 2M** in cost savings.

Full-Stack AI & IIoT Development

- **Predictive Maintenance Dashboard (Angular + ML APIs):** Built Angular (TypeScript) dashboards with state-managed workflows and real-time anomaly tracking, integrating RESTful APIs with containerized ML inference services to enable operational decision support for manufacturing engineers.
- **IIoT Edge Monitoring Systems:** Architected Linux-based IoT edge systems for low-latency data capture, managing fleet deployments via **Ansible** and enabling resilient, real-time data ingestion into cloud ML pipelines.
- **LLM-Enhanced Development (Claude Opus 4.6, Gemini Pro 3):** Leveraged LLMs to refactor code, optimize API pipelines, and auto-generate structured UI components, boosting full-stack development velocity while maintaining production-grade code quality.

Angsana Technology, Software Engineer, Singapore

Aug 2022 – Jul 2023

- Collaborated as part of the core development team to build and deliver a project aligned with Maritime and Port Authority Singapore (MPA) specifications, facilitating the **digitalization of bunkering operations**. (*TypeScript, Python*)
- Spearheaded the implementation of a continuous integration pipeline, automating API and end-to-end tests using GitHub Actions to ensure seamless code commits. (*DevOps, GitHub Actions, Playwright*)

SKILLS

Languages

English · Chinese

Programming Languages

Python · JavaScript · TypeScript · SQL · HTML · CSS

Libraries and Frameworks

Pandas · Numpy · scikit-learn · OpenCV · Ultralytics · NodeJS · ReactJS

Data Engineering & Cloud

Google Cloud Platforms (*Professional Cloud Architect certified*) · Docker · Ansible · Linux

PROJECTS

Theta Agent | Autonomous Stock & Options Analysis Agent

May 2026 – Present

- Built an **autonomous financial research agent** on Claude using **native tool use / function calling** to pull live market data and options chains via yfinance, driving multi-step reasoning toward concrete options strategies.
- Designed a persistent memory layer (JSON/markdown) to retain past research sessions and current positions, giving the agent continuity across runs and cutting redundant analysis.

Applied DSPy RAG Optimization | HDB BTO Guidelines

Jan 2026 – Feb 2026

- Engineered a documentation-grounded RAG pipeline for HDB BTO policy guidelines, integrating **recursive web scraping**, **semantic chunking**, and **DSPy prompt-program optimization (BootstrapFewShot, MIPROv2)** to deliver robust, retrieval-aware responses.
- Generated **QA datasets** using **open-source LLMs (Qwen)** and leveraged **Google's Anti-Gravity LLM-assisted IDE** to accelerate prompt iteration, retrieval debugging, and evaluation-driven optimization.

Shao Kiat's Blog | Self-hosted Blog

Apr 2022 – Present

- Designed and self-hosted a technical blog using **Docusaurus**, covering topics such as **AI, Web3 fundamentals, and data-driven systems**.
- Regularly translate complex technical concepts into clear, beginner-friendly explanations, demonstrating strong written communication skills.

EDUCATION

National University of Singapore

Bachelor of Computing in **Computer Science**, *Second Class (Upper) Honors*

CERTIFICATIONS

Google Cloud Platforms

- **Google Cloud Professional Cloud Architect** – *Issued: Jun 2026*

Coursera

- Smart Analytics, Machine Learning, and AI on Google Cloud – *Issued: Aug 2023*

LEADERSHIP EXPERIENCE & ACTIVITIES

AI Explorer Program, Volunteer Educator (Master Trainer)

Mar 2026 – Present

- In collaboration with **AI Singapore** to design and deliver curriculum on fair and responsible use of AI, with a focus on equipping underprivileged communities in Singapore with practical AI literacy skills to navigate an increasingly AI-driven world.

Micron Transport Committee, Sub-Committee Member

Jul 2023 – Present

- Actively participated in the planning and facilitation of company transport logistics, ensuring smooth transportation services for over 2,000 employees.